

Report: MLLM

Introduction

In this assignment, we are implementing the the SophiaVL-R1 and InternVL3 visual language models with a Lora layer. We will then fine tune the LoRA layer on the MathVista dataset.

The MathVista dataset is a consolidated Mathematical reasoning benchmark within visual contexts. It consists of three datasets: IQTest, FunctionQA and PaperQA. These datasets are tailored to different ways of evaluating logical reasoning over various domains. It contains a mixture of text and images.

InternVL3 is a large multimodal model meaning that it processes text, and images. The model that we will be using in this task is the 8B parameter version of this model. SophiaVL-R1 is another large multimodal model. The one we are using has 3B parameters. Both of these models will have a Lora layer added over them in order to fine-tune the models for a specific dataset without destroying the training and knowledge that was already accumulated by these models. Lora fine-tuning also reduces training time drastically.

Results and Insights

On 3 Epochs, SophiaVL took about 1 hour to train whereas InternVL took less than 10 minutes. The results were about the same on classification. The amount of VRAM for both models are significant. However it since SophiaVL took much more time to fine-tune, InternVL would be the preferred model for use. You can view the results in the included files.